

Cover Letter: Graduate Data Scientist

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Talent Acquisition Team
Bending Spoons

Subject: Application for Graduate Data Scientist

Dear Members of the Talent Acquisition Team,

I am applying for the Graduate Data Scientist position because it combines the work I most enjoy: turning incomplete operational questions into dependable data products and improving the systems that produce them. I recently completed an MSc in Modelling for Science and Engineering at the Universitat Autònoma de Barcelona, specialising in Data Science. My work spans enterprise data and agent products at Aily Labs, scientific machine learning, and end-to-end LLM systems.

At Aily Labs, I took responsibility for integration seams connecting go-to-market data capabilities to SuperAgent, the company's general-purpose, customer-facing LLM reasoning agent. My work progressed from MCP server tools to task-scoped skills that packaged approved instructions, tools, and evaluation criteria around each workflow. I then helped migrate these skills into a governed semantic layer, representing approved SQL tables through curated Markdown resources with their business meaning, relationships, permitted use, and access constraints. Specialised agents could therefore discover and query the right data instead of inferring intent from raw schemas.

I developed and maintained GTM capabilities used by Aily tenants including Sanofi. When an agent's answer quality regressed or a skill, resource mapping, or workflow contract changed, my team was responsible for evaluating and correcting the behaviour from development through production. My contribution was to trace failures across data availability, semantic descriptions, table allowlists, tool selection, DAG payloads, and post-processing, then verify the correction before promotion. The pace required fast intervention without compromising tenant isolation, data correctness, or customer-facing quality.

Changes moved through code review, automated tests, continuous integration, pre-production validation, and controlled production promotion. Question sets declared expected tables, skills, tools, and answer criteria, while traces, deterministic validators, and model-based evaluation checked each result. I also built reporting components that passed analytical outputs, artifact paths, row counts, and execution metadata through explicit contracts to Airflow. This made me responsible for the complete data-product path from analysis and orchestration to validation and consumption.

iTrader is my autonomous trading platform under staged development. Its implemented offline stack learns from OHLCV market windows and solver-specific engineered features rather than hand-authored buy or sell labels. It combines market simulation, validated task specifications, a PPO/GAE solver, task-conditioned FiLM policy and value networks, decomposed rewards for return, drawdown and turnover, proposer backends, and reproducible manifests. Its online design progresses through replay and shadow operation before live execution. Candidate tasks are scored against measured solver capability; deterministic gates choose whether to trade or wait, while risk and execution controls remain authoritative. The goal is continuous improvement without sacrificing reproducibility or safe non-action.

My BSc thesis provides complementary supervised-learning experience. I built and validated a neural network for turbulent-intensity prediction from wind-profile data, connecting feature preparation and model selection to the underlying physical system. Both projects required meaningful targets, separate evaluation evidence, and restraint when an attractive result was not yet reliable.

My MSc thesis demonstrates my ability to integrate an agent into an unfamiliar, distributed ecosystem. I designed and validated a modular ROS 2 and ROS4HRI stack for NAO, separating dialogue, symbolic knowledge, language-model planning, deterministic orchestration, and robot skills. The model proposes plans, while reviewed capabilities and typed execution contracts determine what the robot may do; feedback supports clarification, replanning, or truthful failure. I also built the scenario-based framework used to validate dialogue, knowledge queries, composite tasks, injected failures, recovery, and final reporting. The work treats the LLM, harness, and ROS runtime as one measurable system without obscuring ownership at their interfaces.

Watson is the personal agentic platform I built for my own software development and research. Hermes provides the agent harness and tool workflow, LiteLLM provides model routing and an OpenAI-compatible proxy, llama.cpp serves a quantised Qwen model on the GPU, and ZeroTier makes the endpoint available across my machines. I maintain its launch, health, recovery, and routing paths, benchmarking context size, time to first token, throughput, memory use, and tool-call fidelity before promoting a configuration. Watson accelerates my daily development, but it also makes me operate the serving path behind that value, including observability, recovery, and performance-quality trade-offs.

Bending Spoons appeals to me because the Graduate Data Scientist role places data work inside product and company decisions. The responsibility is not only to analyse data, but to improve its pipelines, define metrics with engineers, and build reporting or forecasting tools that remain useful across products. At Aily, I maintained production-facing data and agent workflows for several consumers. iTrader makes me accountable for a modelling and decision pipeline from raw market data to controlled action. NAO and Watson show that I can integrate AI into unfamiliar systems, build the surrounding software, and evaluate the result instead of trusting plausible output.

I would bring strong Python and SQL foundations, first-principles reasoning, and a high standard of ownership. At ABS Consulting, engineering work on seismic qualification and structural assessment for UK nuclear-power assets reinforced careful validation and reviewable communication. I use AI tools in daily development, but understand their limits because I have built and maintained the data, serving, tool, and evaluation layers around them.

I will continue living in Barcelona after completing my MSc. I maintain permanent residence in the United Kingdom and would welcome sponsorship for a Spain-based role, with flexibility to visit the Madrid office as needed. Thank you for considering my application. I would welcome the opportunity to discuss how my data engineering, applied modelling, and production AI ownership could contribute to Bending Spoons.

Yours sincerely,

Juan David Bendek Williamson

References

1. **Francisco Martin**, Principal Data Scientist, Aily Labs | Manager | francisco.martin@ailylabs.com
2. **Dr. Balogh Miklos**, Professor, BME | BSc Thesis Supervisor | balogh.miklos@gpk.bme.hu
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